Question ID: 483d208d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question

At a state fair, attendees can win tokens that are worth a different number of points depending on the shape. One attendee won S square tokens and C circle tokens worth a total of $1{,}120$ points. The equation $80S + 90C = 1{,}120$ represents this situation. How many more points is a circle token worth than a square token?

Answer

A. 950

B. 90

C. 80

D. 10

Correct Answer: D

Rationale

Choice D is correct. It's given that the equation $80S + 90C = 1{,}120$ represents this situation, where S is the number of square tokens won, C is the number of circle tokens won, and $1{,}120$ is the total number of points the tokens are worth. It follows that $80S$ represents the total number of points the square tokens are worth. Therefore, each square token is worth 80 points. It also follows that $90C$ represents the total number of points the circle tokens are worth. Therefore, each circle token is worth 90 points. Since a circle token is worth 90 points and a square token is worth 80 points, then a circle token is worth $90 - 80$, or 10, more points than a square token.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the number of points a circle token is worth.

Choice C is incorrect. This is the number of points a square token is worth.